Generalization of a Gamma Function Limit

Theorem 1. *For real parameters a, b, c with $c > 0$, the following limit holds:*

$$\lim_{x \to 0^+} \left(\frac{\Gamma\left(\frac{1}{x} + c + ax\right)}{\Gamma\left(\frac{1}{x} + c\right)} x^{ax} \right)^{\frac{b}{x^2}} = e^{ab\left(c - \frac{1}{2}\right)} \tag{1}$$

Proof. We begin by applying the substitution $t = \frac{1}{x}$. As $x \to 0^+$, we have $t \to \infty$. The limit transforms into:

$$\mathcal{L} = \lim_{t \to \infty} \left(\frac{\Gamma\left(t + c + \frac{a}{t}\right)}{\Gamma(t + c)} t^{-\frac{a}{t}} \right)^{bt^2} \tag{2}$$

Let $I(t)$ denote the expression inside the outer parentheses. Taking the natural logarithm yields:

$$\ln I(t) = \ln \Gamma\left(t + c + \frac{a}{t}\right) - \ln \Gamma(t + c) - \frac{a}{t} \ln t \tag{3}$$

Since $t \to \infty$, the term $\frac{a}{t} \to 0$. We expand $\ln \Gamma\left(t + c + \frac{a}{t}\right)$ using a Taylor series centered at $z = t + c$:

$$\ln \Gamma\left(z + \frac{a}{t}\right) = \ln \Gamma(z) + \frac{a}{t}\psi(z) + \frac{a^2}{2t^2}\psi'(z) + \mathcal{O}(t^{-3}) \tag{4}$$

where $\psi(z)$ and $\psi'(z)$ are the digamma and trigamma functions.

We utilize the standard asymptotic expansions for $z \to \infty$:

$$\psi(z) = \ln z - \frac{1}{2z} + \mathcal{O}(z^{-2}) \tag{5}$$

$$\psi'(z) = \frac{1}{z} + \mathcal{O}(z^{-2}) \tag{6}$$

Evaluating at $z = t + c$, we expand with respect to t:

$$\ln(t + c) = \ln t + \frac{c}{t} + \mathcal{O}(t^{-2}) \tag{7}$$

$$\frac{1}{t + c} = \frac{1}{t} + \mathcal{O}(t^{-2}) \tag{8}$$

This gives the expanded polygamma terms:

$$\psi(t + c) = \ln t + \frac{c - 1/2}{t} + \mathcal{O}(t^{-2}) \tag{9}$$

$$\psi'(t + c) = \frac{1}{t} + \mathcal{O}(t^{-2}) \tag{10}$$

Inserting these into $\ln I(t)$:

$$\ln I(t) = \frac{a}{t}\left(\ln t + \frac{c - 1/2}{t} + \mathcal{O}(t^{-2})\right) + \frac{a^2}{2t^2}\left(\frac{1}{t}\right) - \frac{a}{t} \ln t \tag{11}$$

Simplifying cancels the logarithmic dependencies:

$$\ln I(t) = \frac{a(c - 1/2)}{t^2} + \mathcal{O}(t^{-3}) \tag{12}$$

Applying the outer exponent bt^2 yields the final argument for the exponential:

$$\lim_{t\to\infty} bt^2 \ln I(t) = \lim_{t\to\infty} bt^2 \left(\frac{a(c - 1/2)}{t^2}\right) = ab\left(c - \frac{1}{2}\right) \tag{13}$$

Exponentiating this result recovers $\mathcal{L} = e^{ab\left(c-\frac{1}{2}\right)}$. $\square$